Programming Assignment 3

by Paz Lieghton

Applied Probability and Statistics. Prof. Lucas Finazzi

Introduction

This report fits a linear model $A(x) = \hat{a}_1 + \hat{a}_2 x$ measurements (x_i, y_i) with $\sigma = 0.3$ on each y_i. Part A finds the best fit parameters and covariance matrix by weighted least squares and checks the fit quality with χ^2. Part B propagates uncertnties to predicted values, showing what breaks when the off diagonal covariance is ignored which is something often overlooked. Part C runs a Monte Carlo simulation to verify the χ^2 distribution. All code uses NumPy, SciPy, and Matplotlib.

1 Linear Least Squares Fit

1.1 Parameters and Covariance

x	2.0	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
y	2.78	3.29	3.29	3.33	3.23	3.69	3.46	3.87	3.62	3.40	3.99

Table 1: Dataset ($\sigma = 0.3$ on all y_i).

With weights $w = 1/\sigma^2$ one defines S, S_x, S_y, S_{xx}, S_{xy} and $\Delta = SS_{xx} - S_x^2$. The estimators from the normal equations are

$$\hat{a}_1 = \frac{S_{xx}\, S_y - S_x\, S_{xy}}{\Delta}, \qquad \hat{a}_2 = \frac{S\, S_{xy} - S_x\, S_y}{\Delta},$$

with covariance matrix $\mathrm{Cov}(\hat{a}_1, \hat{a}_2) = \frac{\sigma^2}{\Delta}\begin{pmatrix} S_{xx} & -S_x \\ -S_x & S \end{pmatrix}$ Applied to Table 1:

$$\hat{a}_1 = 1.45 \pm 0.721, \qquad \hat{a}_2 = 0.799 \pm 0.286, \qquad \mathrm{Cov}(\hat{a}_1, \hat{a}_2) = -0.205.$$

The off-diagonalis Cov $\propto -S_x < 0$: a larger slope forces a smaller intercept to keep the line through the data, thus the parameters are anti-correlated.

1.2 Goodness of Fit

The chi square statistic $\chi^2 = \sum_i (y_i - \hat{a}_1 - \hat{a}_2 x_i)^2/\sigma^2$ with $\nu = n - 2 = 9$ degrees of freedom gives $\chi^2 = 4.64$ and $p = 0.864$.

Since $\chi^2 < \nu$, two interpretations are possible: the model is correct and this sample happened to scatter less than average (supported by $p \gg 0.05$) as discussed in class, or $\sigma = 0.3$ slightly overestimates the true uncertainty. Either way the linear model is not rejected. Setting $\sigma = 0.2$ in the code flips χ^2 above ν, affirming that the declared σ is driving this flipped relation.

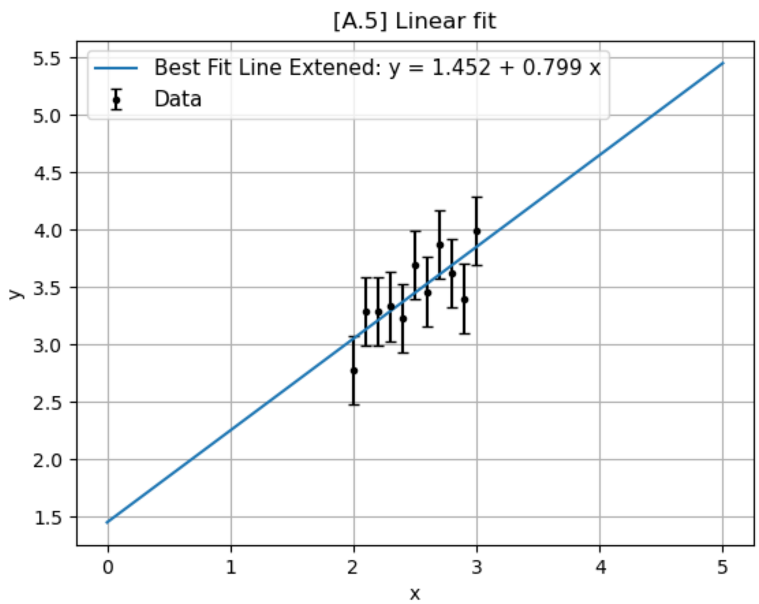

Figure 1: Data with $\pm\sigma$ error bars and best-fit line $\hat{y} = 1.45 + 0.799\,x$ over $x \in [0, 5]$.

2 Uncertainty of Predicted Values and the Role of the Covariance

2.1 Variance and Minimums

For a predicted value $\hat{y}_a = \hat{a}_1 + \hat{a}_2 x_a$, error propagation gives

$$\mathrm{Var}(\hat{y}_a) = \sigma_{a_1}^2 + x_a^2\,\sigma_{a_2}^2 + 2x_a\,\mathrm{Cov}(\hat{a}_1, \hat{a}_2).$$

The cross sterm is negative for all $x_a > 0$, so dropping it (setting $\mathrm{Cov} = 0$) overestimates the uncertainty. Setting it as $d\,\mathrm{Var}/dx_a = 0$:

$$x_a^* = -\frac{\mathrm{Cov}(\hat{a}_1, \hat{a}_2)}{\sigma_{a_2}^2} = \frac{S_x/\Delta}{S/\Delta} = \frac{S_x}{S} = \bar{x}.$$

With uniform weights $\bar{x} = 2.5$, confirmed numerically as $0.205/0.0818 = 2.5$. The line is pinned at the center and uncertainty grows as we get farther from $\bar{x}$.

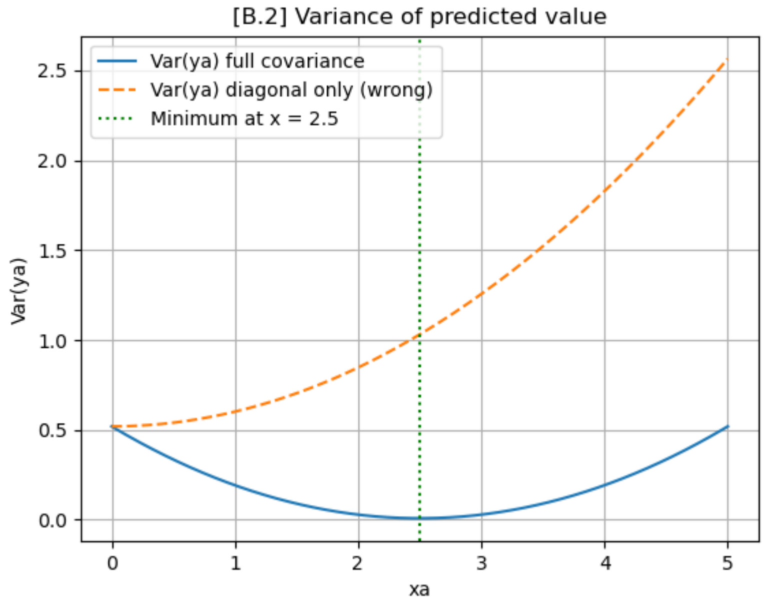

Figure 2: $\mathrm{Var}(\hat{y}_a)$ vs x_a. Blue: full covariance (correct). Orange: diagonal only (incorrect). Green: minimum at $\bar{x} = 2.5$.

2.2 Confidence Bands

Figure 3 shows the $\pm\sqrt{\mathrm{Var}(\hat{y}_a)}$ bands on the fit. The correct band (darker) is narrowest at $\bar{x}$ and widens symmetrically. The diagonal-only band (lighter) is always WIDER. The dashed line marks $\bar{y} = 3.450$, which lies on at the fit at $x = \bar{x}$.

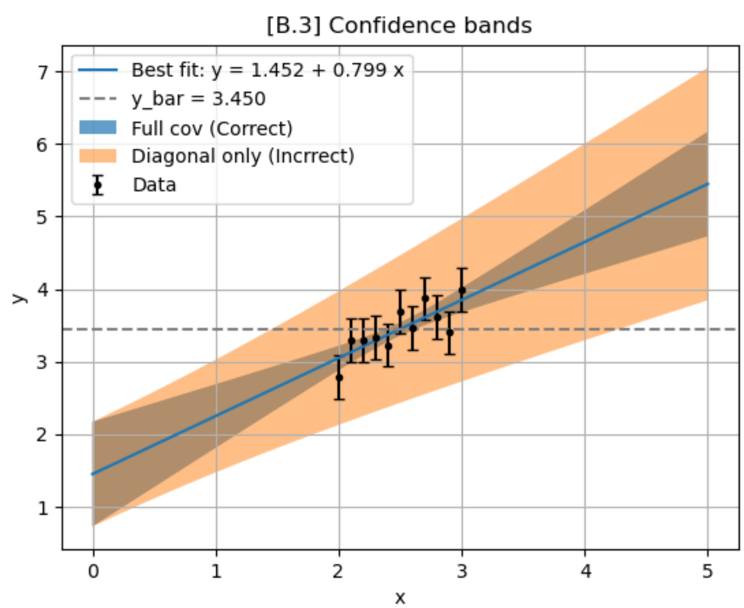

Figure 3: Best fit line with corrct(darker) and diagonal(lighter) confidence bands. Dashed line its at: $\bar{y} = 3.450$.

3 Monte Carlo Validation

$N = 1000$ synthetic datasets were generated by drawing $y_i \sim \mathcal{N}(\hat{a}_1 + \hat{a}_2 x_i,\, \sigma^2)$, fitting each by least squares, and recording the χ^2. The resulting histogram matches the $\chi^2(9)$ density (Figure.4), with simulated mean $8.918 \approx 9$ and std $4.352 \approx 4.24$, confirming the analytic prediction.

If the wrong model were used, the residues would inflate χ^2 above ν and shift the entire histogram to the right, making the misfit immediately visible, in this case its very close.

The bonus C.2 item demonstrates this: see the Appendix for the BONUS excercise, which had a great degree of effort.

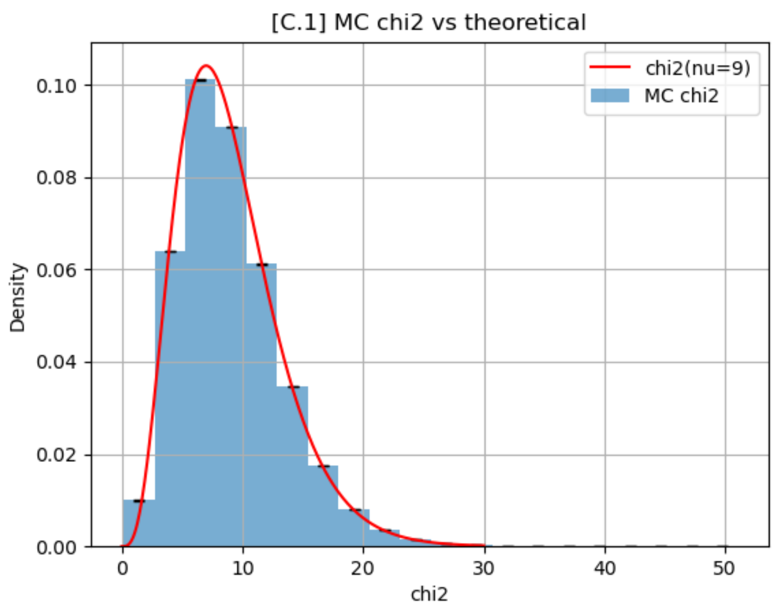

Figure 4: MC χ^2 histogram (1000 datasets) with Poisson errors and theoretical $\chi^2(9)$ the code sent has more bins in this same plot.

Conclusion

Weighted least squares gives the best-fit-line $\hat{y} = 1.45 + 0.799\,x$ with $\chi^2 = 4.64$ ($p = 0.864$), consistent with the linear model. The off diagonal covariance -0.205 encodes the anti correlation between intercept and slope and must be included when propagating uncertainties like the background and motivation of the assignment was: ignoring it overestimates $\mathrm{Var}(\hat{y}_a)$ everywhere, the correct variance is minimised at $\bar{x} = 2.5$. The Monte Carlo simulation confirms that the χ^2 statistic follows $\chi^2(9)$ exactly when the model is good enough.

Sources

- Cowan, Glen. *Statistical Data Analysis in Particle Physics*. Oxford University Press, 1998.
- SciPy Documentation. `scipy.odr` module. `https://docs.scipy.org`
- AI Assistance: Deepseek. helped with the formating and easy access to speed up documentation quirks and plotting (My weakness), plus some formatting for LaTeX
- Python Graph Gallery. Matplotlib examples and reference. `https://python-graph-gallery.com`

A Bonus C.2 Wrong Model Monte Carlo

The same 1000 MC datasets were refitted with a constant $y = c$ (one parameter model like the one treated in class, $\nu_{\text{wrong}} = 10$). Since the true model has slope $\hat{a}_2 = 0.799$, the residuals carry a systematic contribution $a_2(x_i - \bar{x})$ that inflates χ^2.

$$\lambda = \frac{\hat{a}_2^2}{\sigma^2} \sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{(0.799)^2 \times 1.10}{(0.3)^2} \approx 7.80,$$

shifting the expected mean from 10 to $\approx$ 17.8. The MC confirmed this. Figure 5 shows the wrong model histogram clearly displaced from the $\chi^2(10)$ reference exactly it clearly shows that the wrong model displaces itself to the right.

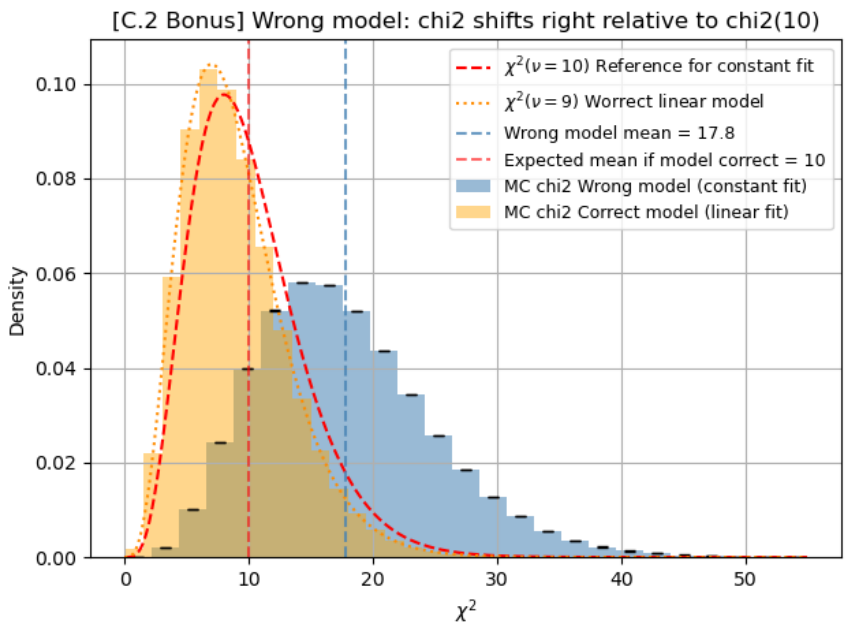

Figure 5: Blue: χ^2 from constant-model fits (wrong). Orange: linear-model fits (correct, from C.1). Red dashed: $\chi^2(10)$ reference. The wrong-model histogram shifts right by $\lambda \approx 7.8$, and its the blue one.